

Further Statistics 1

Notes · Edexcel A Level Further Mathematics — Further Statistics 1 · Chapter 1: Discrete Probability Distributions

Chapter 1: Discrete Probability Distributions

Why this chapter matters

In A Level Maths you learned to use named discrete distributions (Binomial) without ever really being asked to build one from scratch. Further Statistics 1 starts by handing you raw contextual information — a game, a spinner, a faulty machine — and asking you to construct the probability distribution yourself, then find its expectation and variance directly from the definitions. This chapter is the toolkit every later chapter (Poisson, Geometric, PGFs) leans on, so it is worth becoming completely fluent here before moving on.

Concept: what a discrete random variable is

A discrete random variable X takes a countable set of values, each with an associated probability. A probability distribution is a complete listing (as a table or formula) of every value X can take together with $P(X = x)$. Two conditions must always hold:

1. $0 \leq P(X = x) \leq 1$ for every value of x
2. $\sum P(X = x) = 1$, summed over all possible values of x

Many exam questions give a distribution containing an unknown constant (often k), and the first job is to use condition 2 to find it.

Key vocabulary

Term	Meaning
Sample space	The set of all possible outcomes of the underlying experiment
Probability distribution	A table or function giving $P(X = x)$ for every x
Discrete uniform distribution	Every value of X is equally likely: $P(X = x) = 1/n$ if X takes n possible values
Expectation, $E(X)$	The long-run average value of X (the “mean” of the distribution)
Variance, $\text{Var}(X)$	A measure of the spread of X about its mean

Worked example 1: constructing a distribution and finding a constant

EXAMPLE 1.1

Question. A discrete random variable X has probability distribution given by $P(X = x) = kx$ for $x = 1, 2, 3, 4$, and $P(X = x) = 0$ otherwise. Find the value of k .

Step 1. Write down the condition that probabilities sum to 1: $k(1) + k(2) + k(3) + k(4) = 1$. (M1)

Step 2. Simplify: $10k = 1$. (A1)

Step 3. Solve: $k = 1/10$. (A1)

Answer: $k = 0.1$

EXAM TIP

Get into the habit of writing out the full sum before simplifying, even when it looks trivial — it is the single most common place marks are dropped, because it is tempting to jump straight to “ $10k = 1$ ” and if you’ve miscounted terms (e.g. forgotten $x = 4$) the examiner cannot award the method mark. Also note that this technique is identical in spirit to normalising a continuous density function by integrating to 1, which you will meet again in Further Statistics 2 / Core Pure.

Key formula: expectation and variance from first principles

KEY FORMULA

For a discrete random variable X with distribution $P(X = x)$:

- $E(X) = \sum x \cdot P(X = x)$
- $E(X^2) = \sum x^2 \cdot P(X = x)$
- $\text{Var}(X) = E(X^2) - [E(X)]^2$

These formulae are not new — you met them in A Level Maths — but Further Statistics 1 expects you to apply them fluently to distributions you have just built yourself, including ones defined piecewise or by a general algebraic rule, and to functions of X such as $E(aX + b)$ and $\text{Var}(aX + b)$.

It is also worth taking the square root: the standard deviation $\text{SD}(X) = \sqrt{\text{Var}(X)}$ is in the same units as X itself ($\text{Var}(X)$ is in squared units), which is why later chapters — especially the Central Limit Theorem and hypothesis testing — usually quote a standard deviation rather than a variance when describing spread.

Why does $E(X^2) - [E(X)]^2$ measure spread at all? Variance is *defined* as the mean squared distance from the mean, $\text{Var}(X) = E[(X - \mu)^2]$. Expanding the bracket and using $E(aX + b) = aE(X) + b$ term-by-term gives $E[(X - \mu)^2] = E(X^2) - 2\mu E(X) + \mu^2 = E(X^2) - 2\mu^2 + \mu^2 = E(X^2) - \mu^2$. The formula you use in practice is just a shortcut for that definition — it’s algebra, not a new idea.

Key formula: linear transformations

KEY FORMULA

- $E(aX + b) = aE(X) + b$
- $\text{Var}(aX + b) = a^2\text{Var}(X)$

These appear constantly when a question changes the “units” of a problem (e.g. from number of successes to pounds won).

Worked example 2: finding $E(X)$ and $\text{Var}(X)$ from a constructed distribution
EXAMPLE 1.2

Question. A fair spinner is divided into 4 sectors labelled 1, 2, 3, 4. The random variable X is the score obtained on one spin, with $P(X = x) = kx$ for $x = 1, 2, 3, 4$ (as in Example 1, so $k = 0.1$). Find $E(X)$ and $\text{Var}(X)$.

Step 1. Tabulate the distribution: $P(X=1)=0.1$, $P(X=2)=0.2$, $P(X=3)=0.3$, $P(X=4)=0.4$. **(B1)**

Step 2. Compute $E(X) = 1(0.1) + 2(0.2) + 3(0.3) + 4(0.4) = 0.1 + 0.4 + 0.9 + 1.6 = 3$. **(M1A1)**

Step 3. Compute $E(X^2) = 1^2(0.1) + 2^2(0.2) + 3^2(0.3) + 4^2(0.4) = 0.1 + 0.8 + 2.7 + 6.4 = 10$. **(M1A1)**

Step 4. Compute $\text{Var}(X) = E(X^2) - [E(X)]^2 = 10 - 3^2 = 10 - 9 = 1$. **(M1A1)**

Answer: $E(X) = 3$, $\text{Var}(X) = 1$

EXAM TIP

This example reuses the distribution from Example 1 (with k already found) so you can focus purely on the $E(X)/\text{Var}(X)$ mechanics. Use a columnar layout — x , $P(X=x)$, $x \cdot P(X=x)$, $x^2 \cdot P(X=x)$ as adjacent columns — before summing anything. This prevents the arithmetic slips that creep in when the two sums are computed from a jumbled expression, and makes it easy to check work by re-adding a single column.

Worked example 3: a distribution defined by context, plus a linear transformation
EXAMPLE 1.3

Question. A game costs £2 to play. A player rolls a fair six-sided die once. If a 6 is rolled they win £10; if a 4 or 5 is rolled they win £3; otherwise they win nothing. Let W be the player's net profit (winnings minus the £2 cost). Find the probability distribution of W , and hence find $E(W)$ and $\text{Var}(W)$.

Step 1. Identify the possible winnings before deducting the cost: £10 (prob $1/6$), £3 (prob $2/6$), £0 (prob $3/6$). **(M1)**

Step 2. Subtract the £2 cost to get net profit W : $W = 8$ with probability $1/6$; $W = 1$ with probability $2/6$; $W = -2$ with probability $3/6$. **(A1)**

Step 3. Check the distribution sums to 1: $1/6 + 2/6 + 3/6 = 1$ ✓. **(B1)**

Step 4. $E(W) = 8(1/6) + 1(2/6) + (-2)(3/6) = 8/6 + 2/6 - 6/6 = 4/6 = 2/3$. **(M1A1)**

Step 5. $E(W^2) = 64(1/6) + 1(2/6) + 4(3/6) = 64/6 + 2/6 + 12/6 = 78/6 = 13$. **(M1A1)**

Step 6. $\text{Var}(W) = 13 - (2/3)^2 = 13 - 4/9 = 113/9$. **(A1)**

Answer: $E(W) = 2/3$ (the game is favourable to the player, on average, by about 67p per play), $\text{Var}(W) = 113/9 \approx 12.6$

EXAM TIP

This example is deliberately built the way exam questions are: you must first extract a

distribution from a wordy context (translating “wins £10 on a 6” into probabilities) before any statistics can be done. Practise the translation step on its own — many able students who know the $E(X)$ formula perfectly well still lose marks because they built the wrong distribution table. A good habit is to always write “Let X/W be...” and a table before touching any formula.

Procedure: a reliable method for any “construct the distribution” question

1. Identify every distinct outcome (and its associated value of the random variable).
2. Find the probability of each outcome using the context (tree diagrams, sample space diagrams, or combinatorics all still apply).
3. Write the results as a table and check the probabilities sum to 1.
4. Answer whatever is asked (find a constant, find $E(X)/\text{Var}(X)$, find $P(X > a)$, apply a transformation).

Chapter summary

Idea	Key point
Validity of a distribution	Probabilities must be between 0 and 1, and must sum to 1
Constructing a distribution	Translate context into outcomes and probabilities before doing any algebra
$E(X)$	$\sum x \cdot P(X = x)$
$\text{Var}(X)$	$E(X^2) - [E(X)]^2$, where $E(X^2) = \sum x^2 \cdot P(X = x)$
Linear transformations	$E(aX + b) = aE(X) + b$; $\text{Var}(aX + b) = a^2\text{Var}(X)$

Common exam mistakes

- Forgetting a value of x when summing probabilities to find an unknown constant.
- Confusing $E(X^2)$ with $[E(X)]^2$ — these are different quantities and mixing them up is the single most common error in this chapter.
- Applying $\text{Var}(aX + b) = a^2\text{Var}(X) + b$ incorrectly by including b (b has no effect on spread, only on location).
- Not subtracting a fixed cost/entry fee before computing E and Var of a net profit variable.
- Writing a probability greater than 1 or a set of probabilities that doesn't sum to 1 and not noticing.
- Including a value of x outside the stated support (forgetting that $P(X = x) = 0$ everywhere else), which silently drags $E(X)$ or $\text{Var}(X)$ off by including an extra term.

You should now be able to

- Construct a full probability distribution table from a description of a real-world scenario.
- Use $\sum P(X = x) = 1$ to find an unknown constant in a distribution.
- Calculate $E(X)$, $E(X^2)$ and $\text{Var}(X)$ from first principles for any discrete distribution.

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- Apply $E(aX + b)$ and $\text{Var}(aX + b)$ to transformed variables such as net profit in a game, and find a standard deviation as $\text{SD}(X) = \sqrt{\text{Var}(X)}$.